

Lec 3-1 (Mon, Sep 15, 2025)

- chain rule (3.6)
- Implicit differentiation (3.7)
- Linearization and differentials (3.9)
- Proof of the chain rule (3.9)

★★ all locally at a .

finite limit $\Leftarrow$ continuous $\Leftarrow$ differentiable $\Leftarrow$ derivative function f' is continuous at a

$\lim_{x \rightarrow a} f(x)$
exists finitely
(denote by L)

$$\lim_{x \rightarrow a} f(x) = f(a)$$

$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$
exists finitely
(then denote by $f'(a)$)

$$\lim_{x \rightarrow a} f'(x) = f'(a)$$

counter
example

$$f(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

not cont at 0,
limit at 0 NOT exists

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

not diff at 0,
but cont at 0

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

f' not cont at 0,
but f is diff at 0.

$$\begin{aligned} f'(0) &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0} \frac{x^2 \sin(\frac{1}{x}) - 0}{x} = 0 \quad \checkmark \end{aligned}$$

• Chain rule

Q: what is $\left(\sin\left(\frac{1}{x}\right)\right)'$, $x > 0$?

A: Let $f(x) = \frac{1}{x}$ and $g(y) = \sin y$. Then $g \circ f(x) = \sin\left(\frac{1}{x}\right)$.

We are going to answer $(g \circ f)'(x)$.

$$W = h(z) = h(g(f(x)))$$

$$Z = g(y) = g(f(x))$$

Thm. If $y = f(x)$ is differentiable at x and $\underline{z = g(y)}$ is differentiable at $y_0 = f(x_0)$, then $g \circ f$ is differentiable at x_0

and

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

$$\frac{dw}{dx} = \frac{dw}{dz} \cdot \frac{dz}{dy} \cdot \frac{dy}{dx}$$

Rmk: why it is called chain rule:

$$\left. \frac{dz}{dx} \right|_{x=x_0} = \left. \frac{dz}{dy} \right|_{y=f(x_0)} \cdot \left. \frac{dy}{dx} \right|_{x=x_0}$$

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$$

$$\left[\frac{dz}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dw} \cdots \frac{dt}{dx} \right]$$

Pf: We will go back after learning linearization (3.9).

Eg.1. $S(t) = \cos(t^2+1)$. what is $S'(t)$?

$$\begin{aligned} \underline{\text{Sol}'}: S'(t) &= -\sin(t^2+1) \cdot 2t \\ &= -2t \sin(t^2+1) \end{aligned}$$

e.g.2 . $f(x) = x^2 \sin(\frac{1}{x})$, $x > 0$. What is $f'(x)$, $x > 0$?

Pr: $f'(x) = 2x \cdot \sin(\frac{1}{x}) + x^2 [\sin(\frac{1}{x})]'$, $x > 0$.

Let $g(x) = \frac{1}{x}$, $h(y) = \sin y$. $\sin(\frac{1}{x}) = h \circ g(x)$.

$$\therefore [\sin(\frac{1}{x})]' = h'(g(x)) g'(x)$$

$$= \cos(\frac{1}{x}) \cdot (-x^{-2})$$

$$\therefore f'(x) = 2x \cdot \sin(\frac{1}{x}) + x^2 \cos(\frac{1}{x}) \cdot (-x^{-2})$$

$$= 2x \sin(\frac{1}{x}) - \cos(\frac{1}{x}), \quad x > 0.$$

15.

Q: $f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & , x \neq 0 \\ 0 & , x = 0 \end{cases}$ and $f'(0)$.

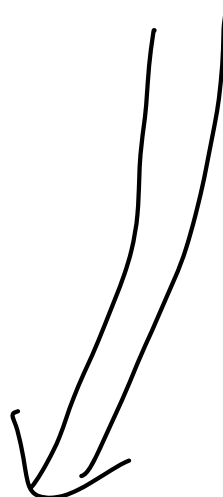
e.g.3 $f(x) = (\frac{x-2}{\cos(x^2)+2})^9$. What is $f'(x)$?

Sol: $f'(x) = 9 \cdot \left(\frac{x-2}{\cos(x^2)+2} \right)^8 \cdot \frac{1 \cdot (\cos(x^2)+2) - (x-2)[- \sin(x^2) \cdot 2x]}{(\cos(x^2)+2)^2}$

=

=

$$\therefore f'(x) = 9 \cdot \left(\frac{x-2}{\cos(x^2)+2} \right)^8 \cdot \frac{\cos(x^2)+2 - (x-2)(-\sin(x^2) \cdot 2x)}{(\cos(x^2)+2)^2}$$

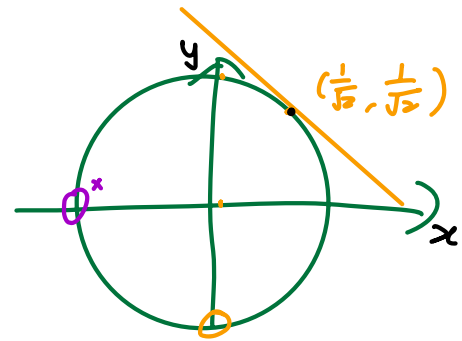


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• Implicit differentiation.

Q: Find the slope of the tangent line to the curve

$$x^2 + y^2 = 1 \text{ at } \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$



A: This one is easier.

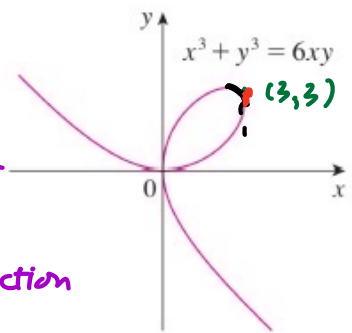
Near the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, the curve is

$$\Rightarrow y(x) = \sqrt{1-x^2}$$

$$\begin{aligned} \therefore \text{slope} &= y'(x) \Big|_{x=\frac{1}{\sqrt{2}}} = \frac{1}{2} (1-x^2)^{-\frac{1}{2}} (-2x) \Big|_{x=\frac{1}{\sqrt{2}}} \\ &= \frac{1}{2} \left(1 - \frac{1}{2}\right)^{-\frac{1}{2}} (-\sqrt{2}) \\ &= \frac{1}{2} \cdot \frac{1}{\sqrt{\frac{1}{2}}} \cdot (-\sqrt{2}) \\ &= -1. \end{aligned}$$

□

Q: Find the slope of the tangent line to the curve $x^3 + y^3 = 6xy$ at point $(3, 3)$.



A: we cannot write $y = h(x)$ explicitly from the above eq.

• But from the curve, locally y can be seen as a function of x near $(3, 3) \Rightarrow$ we write $y = h(x)$, s.t.

Step 1: locally at $(3, 3)$, we write $y = h(x)$:

$$x^3 + h(x)^3 = 6x \cdot h(x), \text{ locally at } x=3.$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dh(x)}{dx} \\ &= h'(x) \big|_{x=3} \end{aligned}$$

Step 2: Take the derivative of $h(x)$.

$$3x^2 + 3h(x)^2 \cdot h'(x) = 6[h(x) + xh'(x)], \text{ locally at } x=3.$$

Step 3: collect all the terms of $h'(x)$:

$$h'(x) = \frac{6h(x) - 3x^2}{3h(x)^2 - 6x} = \frac{2h(x) - x^2}{h(x)^2 - 2x}, \text{ locally at } x=3.$$

$$\Rightarrow h'(x) = \frac{2h(x) - x^2}{h(x)^2 - 2x}, \text{ locally at } 3$$

Step 4: Substitute!

We know $y = h(x)$. Then $h(3) = 3$.

We would like to substitute and get

$$\begin{aligned} h'(x) \big|_{x=3} &= \frac{2h(x) - x^2}{h(x)^2 - 2x} \bigg|_{x=3} = \frac{2 \cdot h(3) - 3^2}{h(3)^2 - 2 \cdot 3} = \frac{2 \cdot 3 - 9}{9 - 6} \\ &= -1. \end{aligned}$$

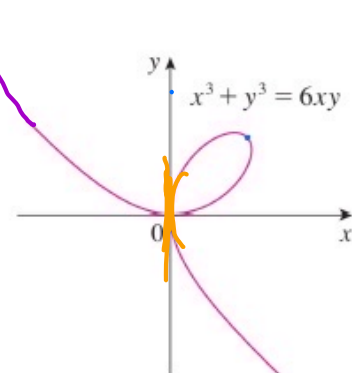
Rmk: In order to make the above approach work, y has to be a function of x at the required point locally at least.

It cannot have a vertical tangent, e.g., it does not work at

$$(x, y) = (0, 0).$$

It is because no matter how small you shrink the curve at

$(0, 0)$, y cannot be seen as a function of x .



Rmk. In fact, we can do it conversely. If we just use the method above and come out with a finite value of $\left. \frac{dy}{dx} \right|_{x=x_0}$, this means the tangent line is not vertical and there is no problem in the procedure.

e.g. Given the curve $2x^3 - 3y^2 = 8$.

What is the slope at $(2, \sqrt{\frac{8}{3}})$?

Step 1. Assume y can be written as a function locally at $(x, y) = (2, \sqrt{\frac{8}{3}})$ and write $y = h(x)$.

Then $2x^3 - 3h(x)^2 = 8$ locally at $(x, y) = (2, \sqrt{\frac{8}{3}})$

Step 2. Take derivative!

$$2 \cdot 3x^2 - 3 \cdot 2h(x)h'(x) = 0 \text{ locally at } (x, y) = (2, \sqrt{\frac{8}{3}})$$

Step 3: Collect all the terms of $h'(x)$!

$$h'(x) = \frac{6x^2}{6h(x)} = \frac{x^2}{h(x)} \text{ locally at } (x, y) = (2, \sqrt{\frac{8}{3}})$$

Step 4: Substitute!

$$h'(2) = \frac{x^2}{h(x)} \Big|_{(x, y) = (2, \sqrt{\frac{8}{3}})} = \frac{2^2}{\sqrt{\frac{8}{3}}} = \sqrt{6}.$$

$\therefore$ the slope at $(2, \sqrt{\frac{8}{3}})$ of the curve is $\sqrt{6}$. (4)

The procedure for finding higher-order derivatives for an implicit function is similar.

e.g. Given the curve $2x^3 - 3y^2 = 8$. What is $\frac{d^2}{dx^2} y$ at the point $(2, \sqrt{\frac{8}{3}})$?

Sol': Step 1. Assume y can be written as a function of x

and write $y = h(x)$. Then

$$2x^3 - 3h(x)^2 = 8 \quad \text{locally at } (x, y) = (2, \sqrt{\frac{8}{3}})$$

Step 2. Take the derivative.

$$x^2 - h(x)h'(x) = 0 \quad \text{locally at } (x, y) = (2, \sqrt{\frac{8}{3}})$$

Once more:

$$2x - h'(x)h'(x) - h(x)h''(x) = 0 \quad \text{locally at } (x, y) = (2, \sqrt{\frac{8}{3}})$$

$$\Rightarrow h''(x) = \frac{2x - h'(x)^2}{h(x)} \quad \text{locally at } (x, y) = (2, \sqrt{\frac{8}{3}})$$

$$\Rightarrow h''(2) = \frac{2 \cdot 2 - h'(2)^2}{h(2)} = \frac{4 - (\sqrt{6})^2}{\sqrt{\frac{8}{3}}}$$

$$= -\sqrt{\frac{3}{2}}.$$

$$\therefore \frac{d^2y}{dx^2} \Big|_{x=2} = h''(2) = -\sqrt{\frac{3}{2}}. \quad \square$$

Linearization.

Given a general function $f(x)$ and a fixed x_0 , it may be

difficult to compute $f(x_0)$, e.g., $f(x) = x^{\frac{1}{3}}$, $x_0 = 27.4$,

to compute $f(x_0) = \sqrt[3]{27.4}$??

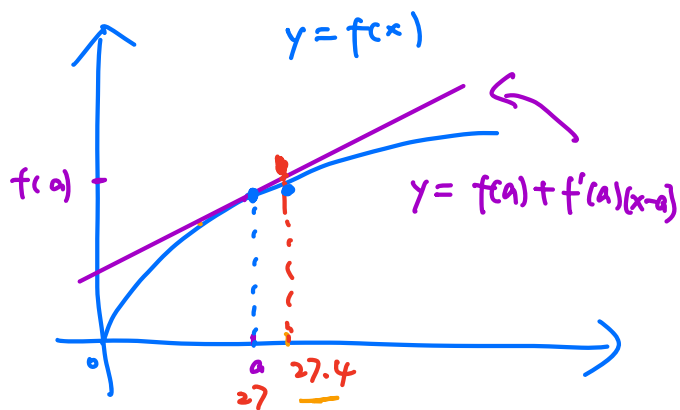
Q: What could be a way to approximate $f(x_0)$?

A: Use tangent line

[They are linear and hence simple!]

Step 1. Assume f is diff' at a .

We know the slope is $f'(a)$.



Step 2. The formula of a straight line passing $(a, f(a))$:

$$y = f(a) + f'(a) \cdot (x-a).$$

$$f'(x) = \frac{1}{3} x^{-\frac{2}{3}}$$

a linear function of x

Step 3. Locally at a , we have: $f(x) \approx f(a) + f'(a) \cdot (x-a)$

$$27^{\frac{1}{3}} + \frac{1}{3} (27)^{-\frac{2}{3}} (x-27) \Big|_{x=27.4}$$

Def. If f is differentiable at $x=a$, then the approximating function

$$L(x) \triangleq f(a) + f'(a) \cdot (x-a)$$

is the linearization of f at a . The approximation

$$f(x) \approx L(x)$$

of f by L is the standard linear approximation of f at a .

The point $x=a$ is the center of the approximation.

ex. 4. Approximate the numerical value of $\sqrt[3]{27.4}$ using standard linear approximation with center 27.

Sol': Step 1: Write the function and take the derivative :

$$f(x) = x^{\frac{1}{3}}$$

$$f'(x) = \frac{1}{3} x^{-\frac{2}{3}} \implies f'(27) = \frac{1}{3} (27)^{-\frac{2}{3}} = \frac{1}{27}$$

Step 2. Write down the straight line passing $(27, \sqrt[3]{27})$.

$$\begin{aligned} \text{Then } L(x) &= f(27) + f'(27)(x-27) \\ &= \sqrt[3]{27} + \frac{1}{27}(x-27) \end{aligned}$$

a linear function,
↙ simple and more friendly
than $\sqrt[3]{x}$

Step 3. Approximate locally: $f(x) \approx L(x)$ locally at 27.

$$\begin{aligned}\therefore f(27.4) &\approx L(27.4) = 3 + \frac{1}{27} (27.4 - 27) \\ &= \frac{27.4}{27} + 2 \\ &= 3.0148148 \dots\end{aligned}$$

□

[For comparison, the true value is $\sqrt[3]{27.4} = 3.01474225$, error < 0.0001]

[A good approximation]

e.g.s. Approximate $(1+\varepsilon)^k \approx 1+k\varepsilon$ for ε (which could be negative or positive with $|\varepsilon|$ small) using standard linear approximation.

Sol': Step 1: write the function and take the derivative!

Let $f(x) = x^k$ and $a = 1$. We have $f'(x) = k \cdot x^{k-1} \Rightarrow f'(1) = k$.

Step 2. write the straight line passing through $(1, f(1))$.

$$L(x) = f(1) + f'(1)(x - 1) = 1^k + k(x - 1).$$

Step 3: Approximate locally: $f(x) \approx L(x)$ locally at 1

$$f(1+\varepsilon) \approx L(1+\varepsilon) = 1 + k(1+\varepsilon - 1) = 1 + k\varepsilon.$$

□

In particular,

$$\sqrt{1.01} = 1.01^{\frac{1}{2}} \approx 1 + \frac{1}{2} \cdot 0.01 = 1.005$$

[true value $\sqrt{1.01} = 1.00498756 \dots$]

$$(1 + \underbrace{5x^4}_{\varepsilon})^{\frac{1}{3}} \approx 1 + (5x^4) \cdot \frac{1}{3} = 1 + \frac{5}{3}x^4 \quad \text{when } 5x^4 \text{ is small}$$

(i.e., when $|x|$ is small).

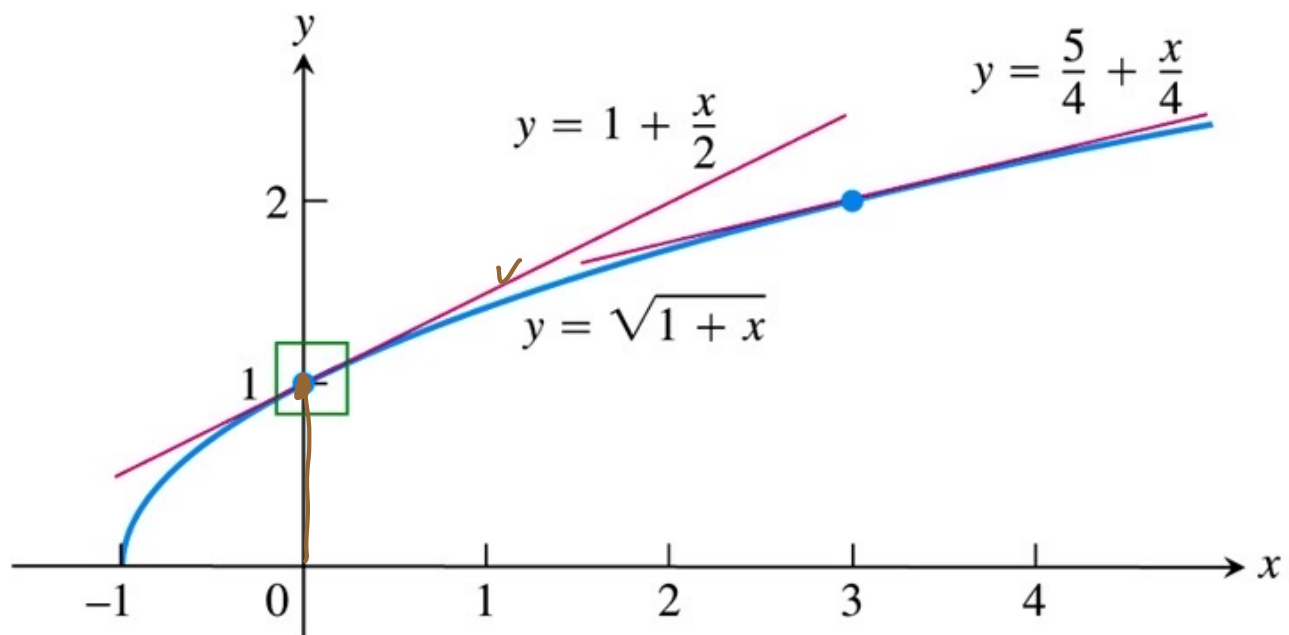


FIGURE 3.40 The graph of $y = \sqrt{1+x}$ and its linearizations at $x = 0$ and $x = 3$. Figure 3.41 shows a magnified view of the small window about 1 on the y-axis.

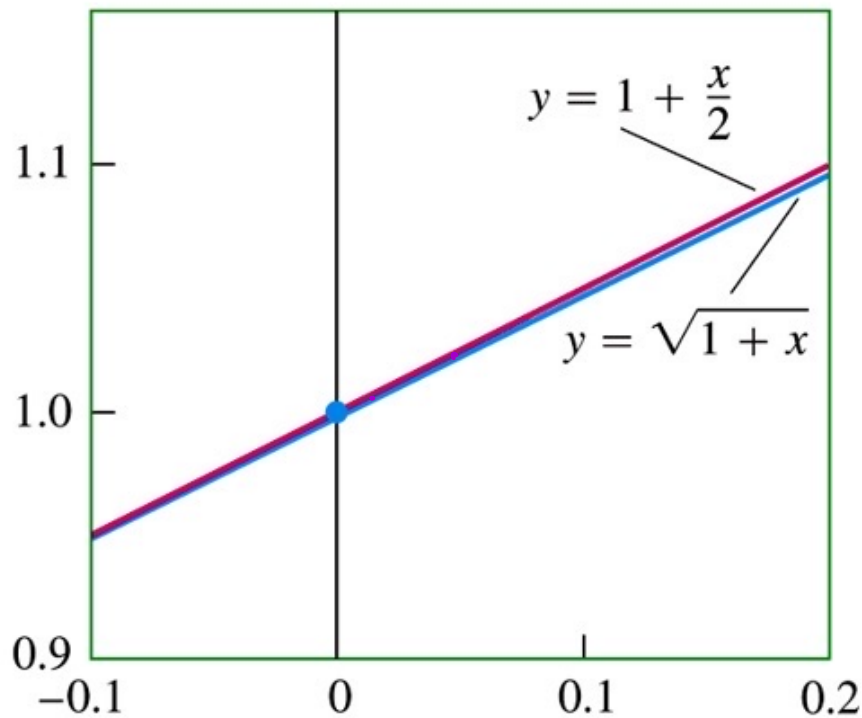


FIGURE 3.41 Magnified view of the window in Figure 3.40.

• Differentials

Def : Let $y = f(x)$ be a differentiable function.

The differential dx is an independent variable.

The differential dy is $dy = f'(x) dx$.

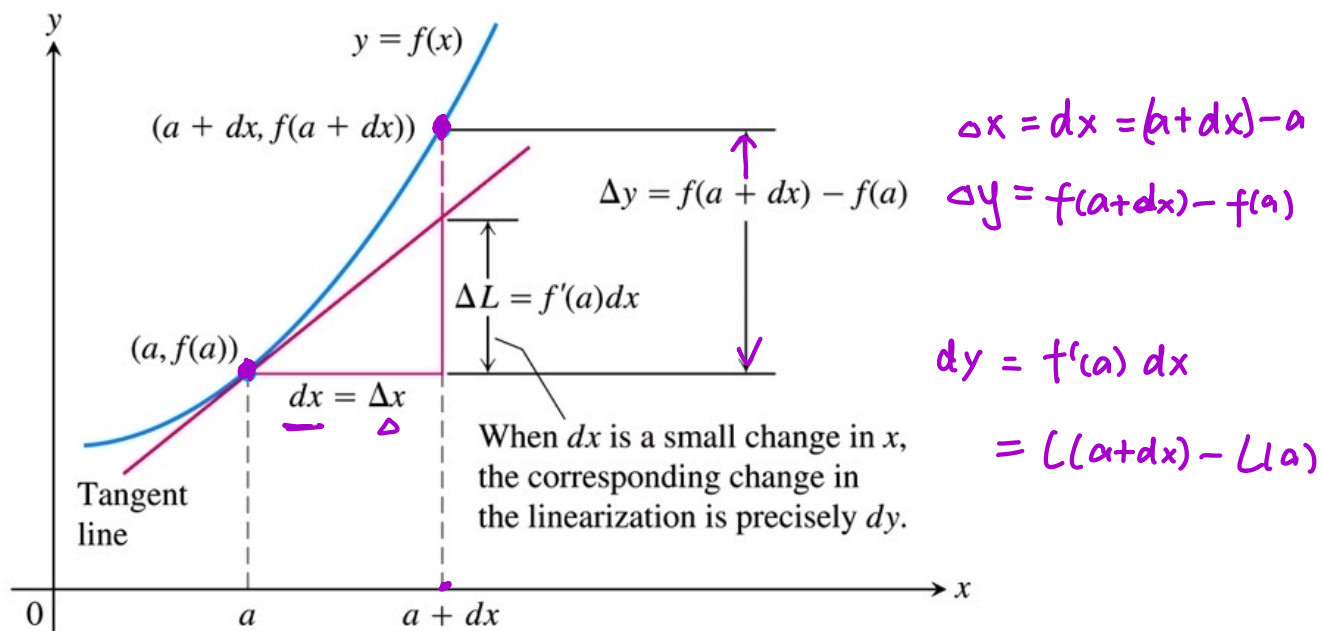


FIGURE 3.43 Geometrically, the differential dy is the change ΔL in the linearization of f when $x = a$ changes by an amount $dx = \Delta x$.

• Geometric meaning:

1. $dx = \Delta x = \text{change in } x\text{-value from } x = a.$

2. $\Delta y = \text{change in } y\text{-value of } f \text{ at } x = a.$

$$= f(a + \Delta x) - f(a)$$

3. $dy = \text{change in } y\text{-value for } L \text{ at } x = a. [L \text{ is linear approximation}]$

$$= L(a + \Delta x) - L(a) = f'(a) dx \cdot \begin{pmatrix} L(a + \Delta x) - L(a) \\ = L(a) + f'(a) \cdot (a + \Delta x - a) - L(a) \\ = f'(a) \Delta x \end{pmatrix}$$

4. Both Δy and dy depend on both a and Δx .

Rmk. ① Differentials can be used to approximate changes:

e.g. If the radius of a circle increases from 10 to 10.022,
then what is the change of area?

Sol': The true change of area is

$$\Delta y = \pi (10.022)^2 - \pi 10^2$$

But if we want to quickly compute the change of area,

we use approximation:

$$\begin{aligned} dy &= f'(a) \cdot dx \Big|_{a=10} \\ &= f'(a) (10.022 - 10) \quad \left(f(x) = \pi x^2 \right) \\ &= 2\pi a \cdot (10.022 - 10) \Big|_{a=10} \\ &= 20\pi \cdot 0.022 \\ &= \underline{0.44\pi} \end{aligned}$$

□.

② we can treat $\frac{dy}{dx}$ as fractions:

$$y = f(x) \Rightarrow \frac{dy}{dx} = f'(x) \Rightarrow dy = f'(x) dx$$

or

$$\Rightarrow \frac{df}{dx} = f'(x) \Rightarrow df = f'(x) dx$$

[dy and df have the same meaning]

. error of standard linear approximation.

Proposition. Fix $x = x_0$. If f is differentiable at x_0 , then we define

$$E(x_0, \Delta x) \triangleq f(x_0 + \Delta x) - [f(x_0) + f'(x_0) \cdot \Delta x]$$

We have

error of standard linear approximation

$$f(x) - L(x)$$

$$\lim_{\Delta x \rightarrow 0} \frac{E(x_0, \Delta x)}{\Delta x} = 0 \iff E(x_0, \Delta x) = o(\Delta x).$$

$\frac{?}{0.01} \rightarrow 0$

[This proposition means the error is very small !
 $(1+0.01)^{\frac{1}{2}}$

If $\Delta x = \underline{0.01}$, then $E(x_0, \Delta x)$ should be much smaller than Δx ,

Say $E(x_0, \Delta x) = \underline{0.001}$.

It means that if we approximate in a small local neighborhood of x_0 , the approximation is rather close to the true value !]

$$\begin{aligned} \text{Pf : } \lim_{\Delta x \rightarrow 0} \frac{E(x_0, \Delta x)}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - [f(x_0) + f'(x_0) \cdot \Delta x]}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \left(\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} - f'(x_0) \right) \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} - f'(x_0) \\ &= f'(x_0) - f'(x_0) = 0. \end{aligned}$$

□.

Proof of the Chain Rule : $[g(f(x))]' = g'(f(x)) \cdot f'(x)$.

$$\text{Pf: } \frac{g(f(x)) - g(f(x_0))}{x - x_0} = \frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)} \cdot \frac{f(x) - f(x_0)}{x - x_0}.$$

Write $y = f(x)$, $z = g(y)$. Then ^{let} $y_0 = f(x_0)$ and $z_0 = g(f(x_0))$.

Use the error formula of standard linear approximation to f :

$$\textcircled{1} \quad \Delta y = f(x) - f(x_0) = f'(x_0)(x - x_0) + E_1(x_0, x - x_0),$$

$$\text{where } \lim_{\Delta x \rightarrow 0} \frac{E_1(x_0, \Delta x)}{\Delta x} = 0.$$

$$\textcircled{2} \quad \Delta z = g(y) - g(y_0) = g'(y_0)(y - y_0) + E_2(y_0, y - y_0),$$

$$\text{where } \lim_{\Delta y \rightarrow 0} \frac{E_2(y_0, \Delta y)}{\Delta y} = 0.$$

Then.

$$\frac{g(f(x_0 + \Delta x)) - g(f(x_0))}{x_0 + \Delta x - x_0} \overset{\text{let } y_0 = f(x_0) \text{ and } z_0 = g(f(x_0))}{=} \frac{g(y_0 + \Delta y) - g(y_0)}{\Delta y} \cdot \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \quad \Delta x \rightarrow 0, \Delta y \rightarrow 0$$

$$= \frac{g'(y_0)(y - y_0) + E_2(y_0, y - y_0)}{\Delta y} \cdot \frac{f'(x_0)(x - x_0) + E_1(x_0, x - x_0)}{\Delta x}$$

$$= \left[g'(y_0) + \frac{E_2(y_0, \Delta y)}{\Delta y} \right] \cdot \left[f'(x_0) + \frac{E_1(x_0, \Delta x)}{\Delta x} \right]$$

$$\text{As } \Delta x \rightarrow 0, \Delta y = f(x_0 + \Delta x) - f(x_0) \rightarrow 0,$$

then $\frac{E_2(y_0, \Delta y)}{\Delta y} \rightarrow 0$.

Also $\frac{E_1(x_0, \Delta x)}{\Delta x} \rightarrow 0$.

$$\begin{aligned} \therefore \lim_{\Delta x \rightarrow 0} \frac{g(f(x_0 + \Delta x)) - g(f(x_0))}{x_0 + \Delta x - x_0} &= \lim_{\Delta x \rightarrow 0} \left[g'(y_0) + \frac{E_2(y_0, \Delta y)}{\Delta y} \right] \lim_{\Delta x \rightarrow 0} \left[f'(x_0) + \frac{E_1(x_0, \Delta x)}{\Delta x} \right] \\ &= [g \circ f(x_0)]' = g'(y_0) \cdot f'(x_0) \\ &= g'(f(x_0)) \cdot f'(x_0). \quad \square \end{aligned}$$

$$\begin{aligned} \frac{g(f(x)) - g(f(x_0))}{x - x_0} &= \frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)} \cdot \frac{f(x) - f(x_0)}{x - x_0} \\ &\stackrel{y_0 = f(x_0)}{=} \frac{g'(y_0) \Delta y + \underbrace{E_1(y_0, \Delta y)}_{=0}}{\Delta y} \cdot \frac{f'(x_0) \Delta x + E_2(x_0, \Delta x)}{\Delta x} \\ &= \frac{g'(y_0) \Delta y + \underbrace{O(\Delta y)}_{=0}}{\Delta y} \cdot \frac{f'(x_0) \Delta x + O(\Delta x)}{\Delta x} \end{aligned}$$

$$\begin{aligned} \therefore \lim_{\Delta x \rightarrow 0} \frac{g(f(x)) - g(f(x_0))}{x - x_0} &= \lim_{\Delta x \rightarrow 0} \frac{g'(y_0) \Delta y + O(\Delta y)}{\Delta y} \lim_{\Delta x \rightarrow 0} \frac{f'(x_0) \Delta x + O(\Delta x)}{\Delta x} \\ &= g'(y_0) \cdot f'(x_0). \quad \square \end{aligned}$$

□